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Math 400

Non-Online Homework

The following problems are included as part of your homework. I will not collect these problems but you should still do them. **Problems like these can show up on exams.**

Section 2.2 page 93 # 23, 52, 54

Please include the following problems as part of the Section 2.2 Homework:

A. Let $f(x)=\begin{cases} 2, \text{ if } x \geq 3 \\ -2, \text{ if } x < 3 \end{cases}$, $\quad a = 3$

(a.) Graph the function *f*.

(b.) Use the graph of f to find $\lim_{x \to a^+} f(x)$ and $\lim_{x \to a^-} f(x)$.

(c.) Find $\lim_{x \to a} f(x)$ if it exist. If it does not exist, explain why?

B. Let $f(x)=\begin{cases} \dfrac{x^2-25}{x+5}, \text{ if } x \neq -5 \\ -9 \quad , \text{ if } x = -5 \end{cases}$, $\quad a = -5$

(a.) Graph the function *f*.

(b.) Use the graph of f to find $\lim_{x \to a^+} f(x)$ and $\lim_{x \to a^-} f(x)$.

(c.) Find $\lim_{x \to a} f(x)$ if it exist. If it does not exist, explain why?

Section 2.2 Homework continued on the next page.

Section 2.2 Homework (continued)

C. Let $f(x)=\begin{cases}\sqrt{x-3}\text{ , if } x>3\\ (x-3)^2\text{, if } x\le 3\end{cases}$, $\quad a=3$

(a.) Graph the function *f*.

(b.) Use the graph of f to find $\lim\limits_{x\to a^+} f(x)$ and $\lim\limits_{x\to a^-} f(x)$.

(c.) Find $\lim\limits_{x\to a} f(x)$ if it exist. If it does not exist, explain why?

D. Let $f(x)=\begin{cases}\ln x\text{ , if } x>0\\ \dfrac{1}{x}\text{ , if } x<0\end{cases}$, $\quad a=0$

(a.) Graph the function *f*.

(b.) Use the graph of f to find $\lim\limits_{x\to a^+} f(x)$ and $\lim\limits_{x\to a^-} f(x)$.

(c.) Find $\lim\limits_{x\to a} f(x)$ if it exist. If it does not exist, explain why?

E. Let $f(x)=\tan x$, $\quad a=\dfrac{\pi}{4}$

(a.) Graph the function *f*.

(b.) Use the graph of f to find $\lim\limits_{x\to a^+} f(x)$ and $\lim\limits_{x\to a^-} f(x)$.

(c.) Find $\lim\limits_{x\to a} f(x)$ if it exist. If it does not exist, explain why?

Section 2.3 page 103 # 22, 32, 51, 62, 63

Please include the following problems as part of your Section 2.3 Homework:

A. Find the following limit, if it exists. (**Hint:** Factor the numerator and denominator.)

$$\lim_{x\to\frac{3\pi}{2}} \frac{\sin^2 x + 6\sin x + 5}{\sin^2 x - 1}$$

B. Find the following limit, if it exists. (**Hint:** Use $\sin^2 x + \cos^2 x = 1$ to make a substitution.)

$$\lim_{x\to 0} \frac{\cos x - 1}{\sin^2 x}$$

C. True or False. If the statement is false, then provide a **counterexample**.

Consider a function $f(x)$ with the property that $\lim_{x\to a} f(x) = 0$. Now consider another function $g(x)$ also defined near a. Then $\lim_{x\to a}\left[f(x)g(x)\right] = 0$.

D. If $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$, then which of the following is true about $\lim_{x\to a} \frac{f(x)}{g(x)}$?

Choose the correct answer. Explain why you chose that answer.

(a.) must exist (b.) not enough information (c.) does not exist

Section 2.5 page 124 # 6, 19, 21, 24, 28, 29, 30, 32, 37, 41, 48, 51

Please include the following problems as part of your Section 2.5 Homework:

A. Find the open intervals where the function is continuous.

$$f(x) = \begin{cases} 5x \text{ , if } x < 1 \\ x^2 + 3x, \text{ if } x \geq 1 \end{cases}$$

Section 2.6 page 137 # 9, 36, 57

Please include the following problems as part of your Section 2.6 Homework:

A. Find the following limits, if they exist. Do this analytically.

$$\lim_{x\to\frac{\pi}{12}^{+}} \sec\left(2x+\frac{\pi}{3}\right), \quad \lim_{x\to\frac{\pi}{12}^{-}} \sec\left(2x+\frac{\pi}{3}\right), \quad \lim_{x\to\frac{\pi}{12}} \sec\left(2x+\frac{\pi}{3}\right)$$

B. Find the limit or show that it does not exist.

(a.) $\lim_{x\to\infty} \dfrac{\sin^2 x}{x^2+1}$ (**Hint:** Use squeeze theorem)

(b.) $\lim_{x\to\infty} \cot^{-1}(-\ln x)$ (Do this analytically.)

C. Determine the end behavior of the following function. (Do this algebraically.)

$$f(x)=\frac{\sqrt[4]{16x^8-12x^2}}{7x^2+\sqrt[5]{x^{10}+2x-8}}$$

D. Find the following limits if they exist. Explain. (Do this analytically.)

(a.) $\lim_{x\to-\infty} \cot^{-1}(\ln|10-8x|)$

(b.) $\lim_{x\to\infty} \cos\left[\operatorname{arccsc}\left(\dfrac{x^2+3x+1}{x+2}\right)\right]$

Section 2.4 page 114 # 15, 17, 21, 42

Please include the following problem as part of your Section 2.4 Homework:

A. Use the precise definition of a limit to prove the following limit.

$$\lim_{x\to -4} \frac{1}{(x+4)^2} = \infty$$

Section 2.7 page 148 # 5, 8, 17, 21, 24, 25, 43, 53 (a.), (b.)

Section 2.8 page 162 # 28, 30, 39, 42, 43, 44

Please include the following problems as part of the Section 2.8 Homework:

A. Use the definition, $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$, to find the derivative.

$$f(x) = \frac{-8}{\sqrt{x+1}}$$

B. Let $f(x) = (x-1)^{\frac{1}{3}}$.

(a.) Sketch the graph of f.

(b.) Use the limit definition of a derivative to show that *f* is not differentiable at x = 1.